

Mohammed Matin Shaikh

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Email

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GitHub

Portfolio

Surat, Gujarat

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT)

B.Tech in Computer Science and Engineering

CGPA: 7.89

2024 – 2028

Smt. I. N. Tekrawala High School

Higher Secondary Certificate (HSC)

JEE Main: 99.39 — 85%

2024

EXPERIENCE

Research Intern (Deep Learning)

Under supervision of Dr. Mukesh Zaveri (Profile)

SVNIT (ISRO Funded Project)

Dec 2025 – May 2026

- Built a hybrid CNN + Transformer model for **image super-resolution (4×)** on RGB images.
- Improved reconstruction quality from **25 dB (baseline CNN)** and **26 dB (MIMO)** to **27.3 dB PSNR**.
- Combined convolutional layers with attention blocks to better preserve fine details and capture global context.
- Ongoing work: extending the model to **multi-spectral (MSI)** and **hyperspectral (HSI)** image data.

PROJECTS

MealDesk: Odoo POS Cafe

GitHub

Odoo × Indus Hackathon '26 Finalist (Top 300 out of 2200+)

- Built modular backend with graceful shutdown and connection-pooled database, alongside React Router-optimized frontend to minimize unnecessary API calls in a 24-hour hackathon.
- Designed ultra-strict custom authentication enforcing **1:1 employee-device-tab sessions** using HTTPOnly cookies + tab-scoped IDs with Redis-backed concurrency control and **XSS-resilient session security**.
- Implemented rolling TTL heartbeat mechanism for near real-time session invalidation and prevention of multi-tab/device misuse.
- Integrated **RAG-based assistant** using **Gemini API**, architected with a **decoupled microservice architecture** enabling dynamic company policy responses.
- Integrated Razorpay webhook-based payments using **event-driven architecture**.

Banking System

GitHub Stack: React, Node.js, Express, PostgreSQL, JWT

- Built a secure banking platform with authentication using **JWT** and **bcrypt**, ensuring resilience against **race conditions** and maintaining **ACID-compliant transactions**.
- Developed a responsive **React** dashboard with a **modular backend** and **decoupled database schema**, leveraging **PostgreSQL procedures, triggers, and functions** for robust transaction management.
- Integrated **email notifications** for account actions such as registration, login alerts, and transaction updates.

StudHelp: Campus Club & Event Platform

GitHub — Live Demo

Stack: React, Node.js, Express, PostgreSQL (Neon), Socket.IO

- Built a full-stack campus platform integrating club management, event registration, team coordination, and real-time messaging with role-based access control solving real world issue.
- Developed real-time group and direct messaging using Socket.IO with multi-room support and persistent chat storage.
- Implemented event discovery, scheduling, and user profile management.
- Configured periodic **health-check API pings** via **UptimeRobot** (5-min interval) to prevent backend cold starts and ensure high availability.

SKILLS

- Languages:** C++, Python, JavaScript, SQL, C.
- Web Development:** React.js, Node.js, Express.js, HTML/CSS, WebRTC, Socket.io
- Deep Learning:** PyTorch, CNNs (U-Net), Transformers (ESRT), NumPy, Pandas.
- Generative AI:** RAG, LangChain, LangGraph, MCP.
- Databases:** PostgreSQL, MongoDB, Vector Databases.
- DevOps & Systems:** Docker, CI/CD, Kubernetes.
- Core Concepts:** DSA, OOP, DBMS, CN, TOC, IS.

ACHIEVEMENTS

- Odoo × Indus Hackathon '26 Finalist:** Selected among **top 300** teams from 2200+ registrations for a 24-hour offline hackathon.
- Competitive Programming:** Solved **500+** problems across platforms; achieved **LeetCode 1400+** and active on Codeforces.
- Nexus Web Wonder:** Secured **Top 3** position in a college-level web development competition.

EXTRA-CURRICULAR ACTIVITIES

- **Student Mentor (DSA)** Jan 2026 – Apr 2026
Conducted weekly mentorship sessions for juniors, simplifying core DSA concepts and problem-solving approaches.
- **Executive Member, ACM SVNIT** Aug 2025 – May 2026
Guided juniors and contributed to building a strong coding culture.
- **Member, Nexus (Technical Club)** Aug 2025 – May 2026
Helped juniors with web development and problem-solving, fostering a collaborative environment.